KAIZEN MATHS

Lesson starter

Powers of 10 and Standard Form

Multiplication by Powers of 10

Name _____

S1

$170 \times 10{,}000$

S2

$3 \times 1{,}000$

S3

$509 \times 100{,}000$

S4

$7.06 \times 1{,}000$

Show your working

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NUM-F1AA-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Powers of 10 and Standard Form

Multiplication by Powers of 10

Name _____

E1

$642 \times 1{,}000$

E2

6.43×100

E3

50×100

KAIZEN MATHS

Exit ticket

Powers of 10 and Standard Form

Multiplication by Powers of 10

Name _____

E1

642 × 1,000

E2

6.43 × 100

E3

50 × 100

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Show your working

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Exit ticket

Powers of 10 and Standard Form

Multiplication by Powers of 10

Name _____

E1

642 × 1,000

E2

6.43 × 100

E3

50 × 100

NUM-F1AA-L1 · E

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Multiplication by Powers of 10

Name _____

E1

642 × 1,000

E2

6.43 × 100

E3

50 × 100

NUM-F1AA-L1 · E

Show your working

Teacher answers & working

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S1

$170 \times 10{,}000$

Answer:

1,700,000

Working:

Step 1: Identify the power of 10

We are multiplying by 10,000 = 10^4

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 4 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

170.0 → digits shift 4 places left → 1,700,000.0

Step 3: Write the final answer

$170 \times 10{,}000 = 1{,}700{,}000$

S2

$3 \times 1{,}000$

Answer:

3000

Working:

Step 1: Identify the power of 10

We are multiplying by 1,000 = 10^3

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 3 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

3 → digits shift 3 places left → 3000

Step 3: Write the final answer

$3 \times 1{,}000 = 3000$

Teacher answers & working

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S3

$509 \times 100{,}000$

Answer:

50,900,000

Working:

Step 1: Identify the power of 10

We are multiplying by 100,000 = 10^5

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 5 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

509.0 → digits shift 5 places left → 50,900,000.0

Step 3: Write the final answer

$509 \times 100{,}000 = 50{,}900{,}000$

S4

$7.06 \times 1{,}000$

Answer:

7060

Working:

Step 1: Identify the power of 10

We are multiplying by 1,000 = 10^3

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 3 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

7.06 → digits shift 3 places left → 7060

Step 3: Write the final answer

$7.06 \times 1{,}000 = 7060$

Teacher answers & working

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E1

$642 \times 1{,}000$

Answer:

642,000

Working:

Step 1: Identify the power of 10

We are multiplying by 1,000 = 10^3

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 3 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

642.0 → digits shift 3 places left → 642,000.0

Step 3: Write the final answer

$642 \times 1{,}000 = 642{,}000$

E2

6.43×100

Answer:

643

Working:

Step 1: Identify the power of 10

We are multiplying by 100 = 10^2

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 2 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

6.43 → digits shift 2 places left → 643

Step 3: Write the final answer

$6.43 \times 100 = 643$

Teacher answers & working

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E3

50×100

Answer:

5,000

Working:

Step 1: Identify the power of 10

We are multiplying by 100 = 10^2

Step 2: Shift the digits through the fixed decimal point

Multiplying by a power of 10 shifts the digits 2 places left into larger place-value columns. The decimal point stays fixed between ones and tenths.

50.0 → digits shift 2 places left → 5,000.0

Step 3: Write the final answer

$50 \times 100 = 5{,}000$